Target Oriented Distributionally Robust Optimization and Its Applications to Surgery Allocation

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Paper

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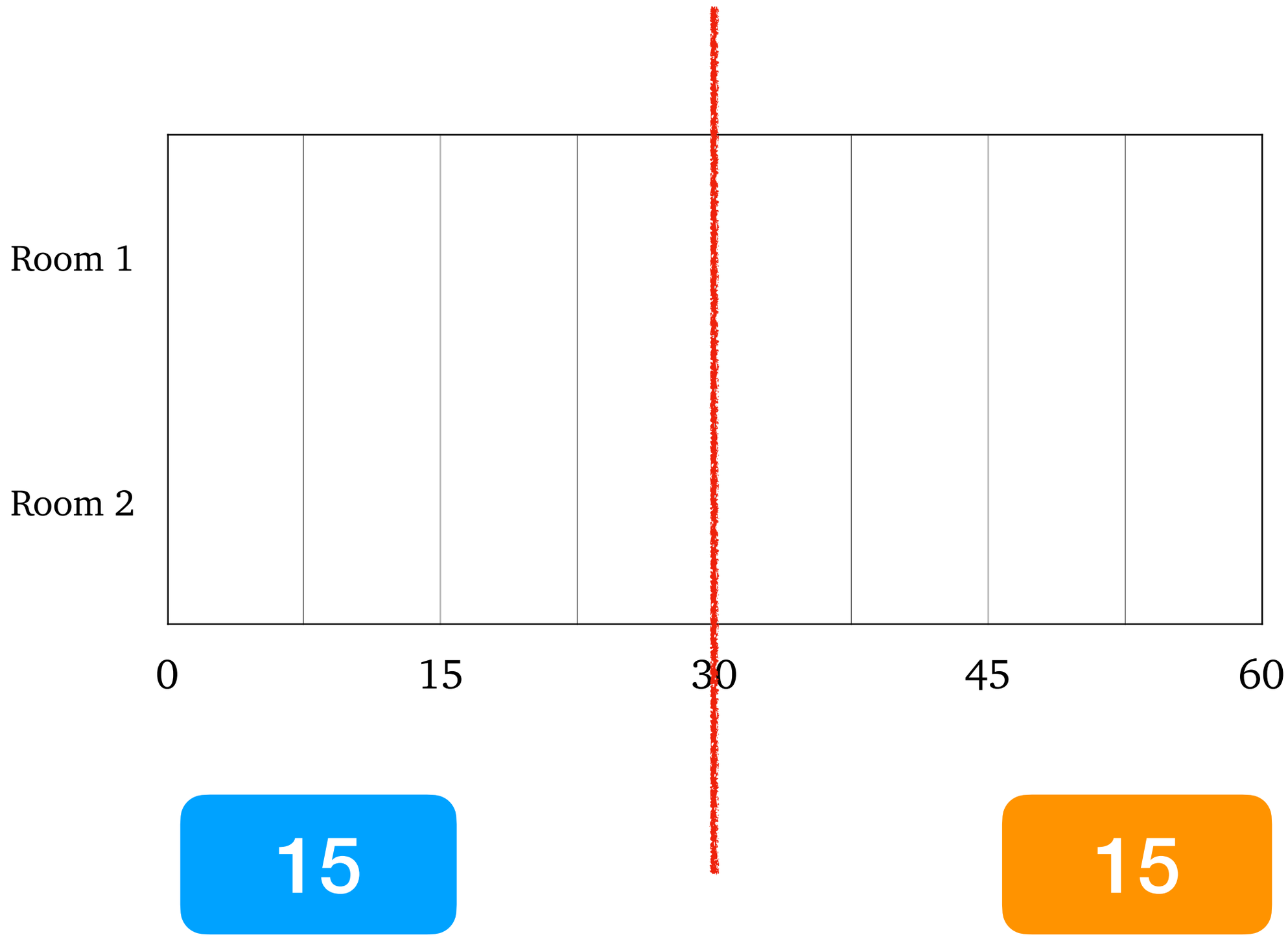
Problem Statement



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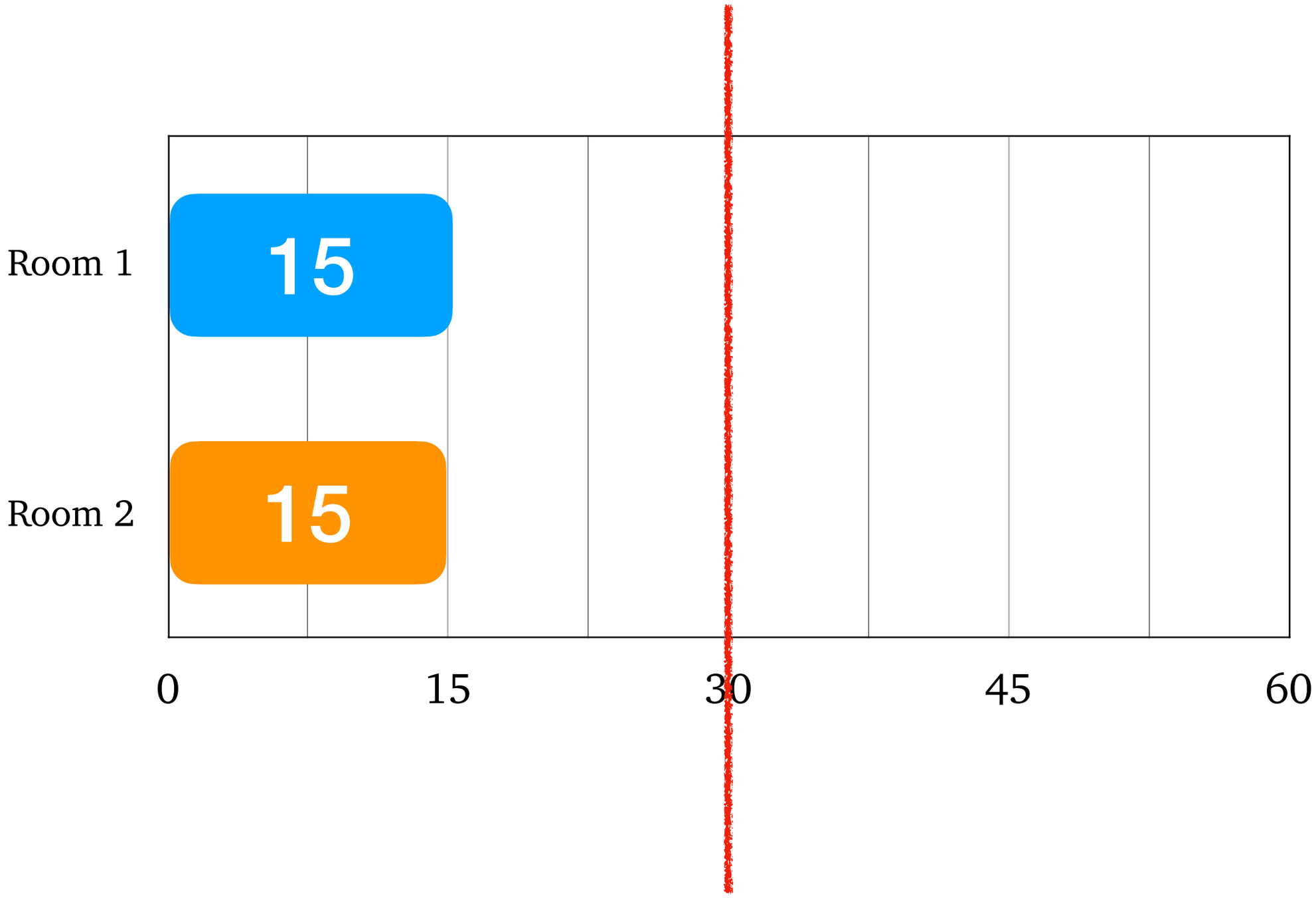

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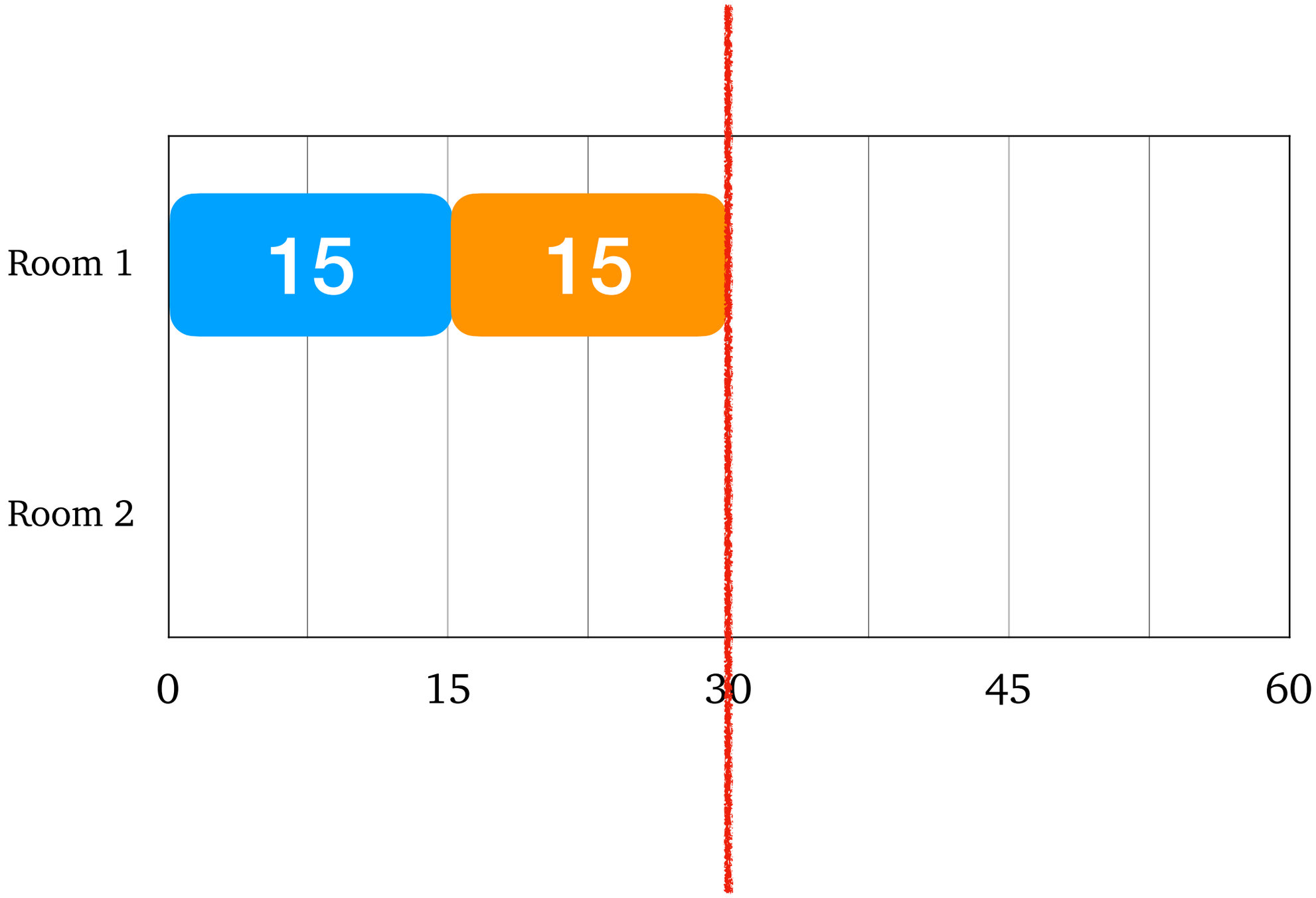

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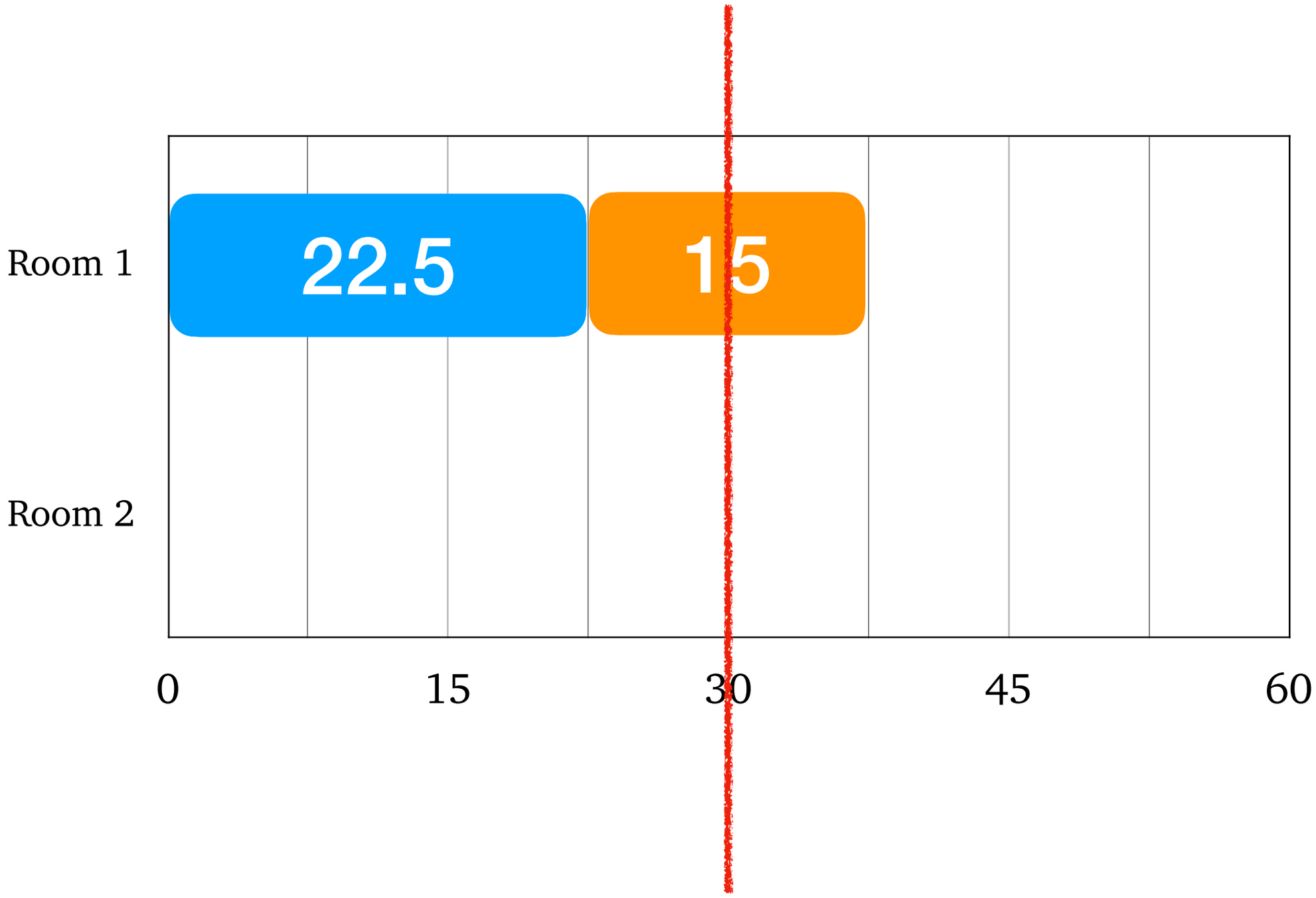

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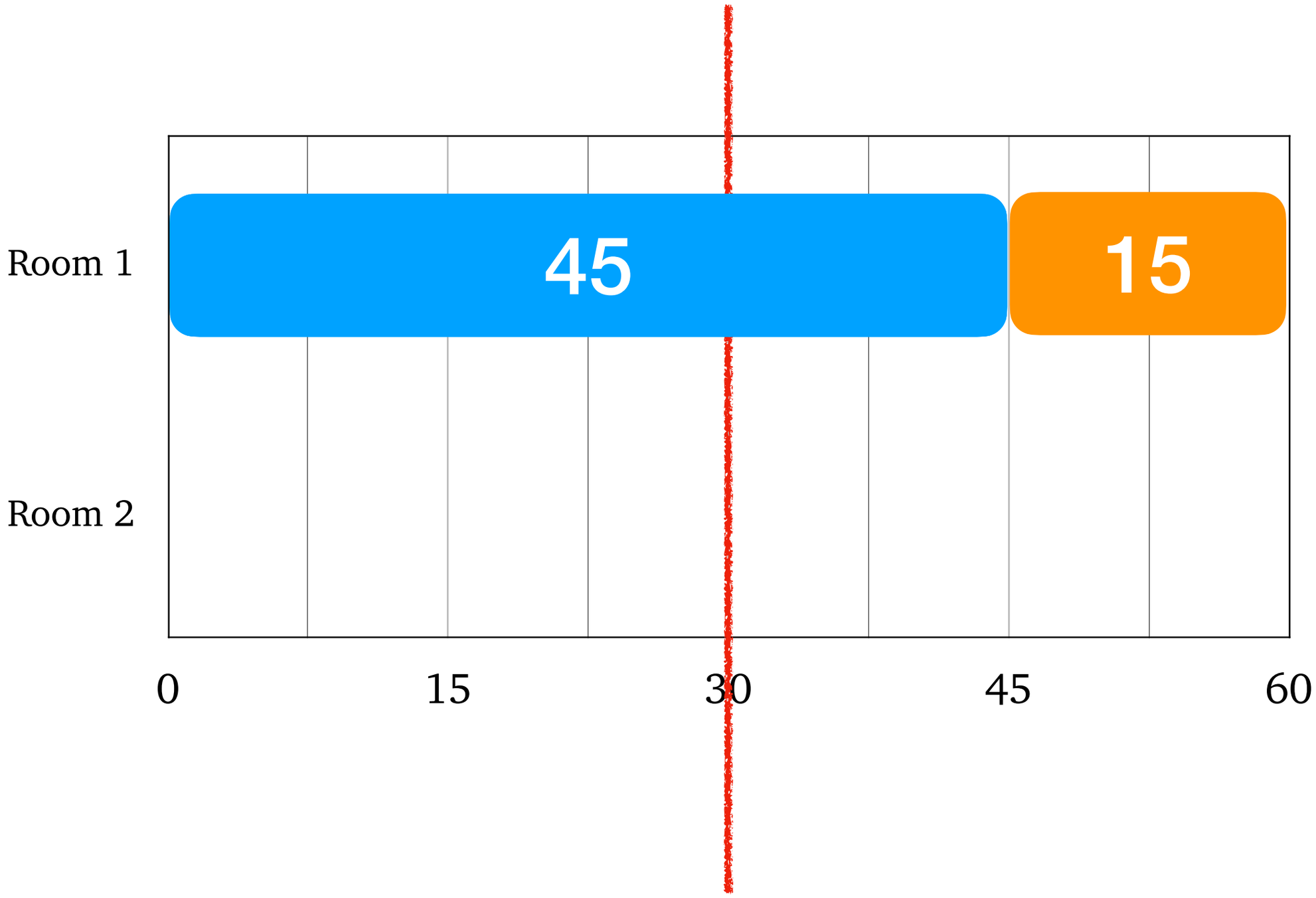

“[…] 12 minutes from the time of call to the arrival of an ambulance […] our target is to meet this response time in 92.5% of all emergency calls.”

–Fire Services Department, HKSARG

95% chance:
Overtime (OT) $\leq$ 10

Problem Statement

OT	Chance
0	95%
15	1%
35	2%
1,000	2%

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- There is no limit on how large the violation will be once it goes beyond the threshold.
- This can be catastrophic in time-sensitive environments.

Problem Statement

90% chance:
OT $\leq$ 5

95% chance:
OT $\leq$ 10

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90% chance:
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95% chance:
OT $\leq$ 10

99% chance:
OT $\leq$ 20

99.5% chance:
OT $\leq$ 50

Problem Statement

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95% chance:
OT $\leq$ 10

99% chance:
OT $\leq$ 20

99.5% chance:
OT $\leq$ 50

99.95% chance:
OT $\leq$ 100

99.99% chance:
OT $\leq$ 1000

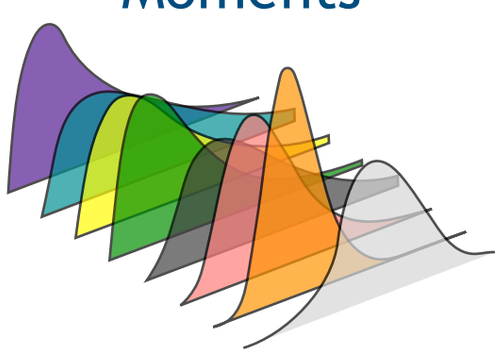
- Not every loophole is covered.
- Finding all probability levels and thresholds is hard.
- Chance-constrained problems are often difficult to solve, not to mention one with many chance constraints.

- Uncertain constraints
- Scalability issue for comprehensive protection

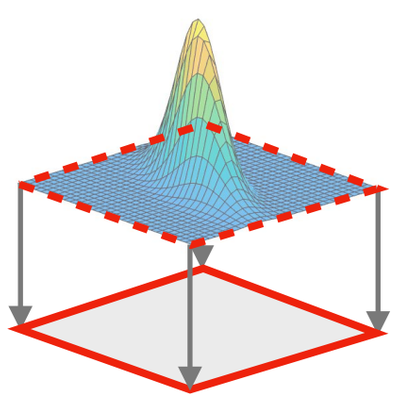
Key Contributions

- Aiming to tackle the two problems, we defined the Coherent Risk Enveloping Measure (CREM).
- Examples and solution methods in various distributionally robust settings are given.

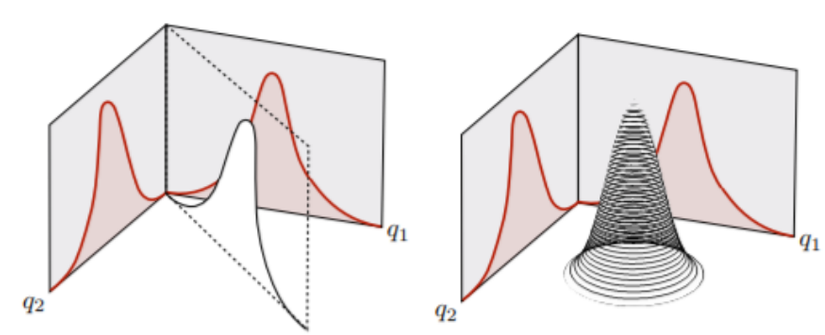
Moments



Support



Structural Information



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- Aiming to tackle the two problems, we defined the Coherent Risk Enveloping Measure (CREM).
- Examples and solution methods in various distributionally robust settings are given.
- Today: Utility-based CREM

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Utility-based CREM

- Pick a function u such that it is
  - Normalized, convex, and non-decreasing
  - $u(w) > u(0) = 1$ for all $w > 0$

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Utility-based CREM

- Pick a function u such that it is
  - Normalized, convex, and non-decreasing
  - $u(w) > u(0) = 1$ for all $w > 0$
- Given a random variable $\tilde{z}$, the utility-based CREM is defined below.

$$\rho_u(\tilde{z}) = \inf\left\{k > 0 \ : \ \mathbb{E}\left[u\left(-\frac{\tilde{z}}{k}\right)\right] \leq 1\right\}$$

A positive realization of $\tilde{z}$ is preferred.

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Corollary 1

Given $\rho^* = \rho_u(\tilde{z})$, we have

$$\text{P(Loss > } \phi\text{)=} \ \mathbb{P}(-\tilde{z} > \phi) \leq \frac{1}{u(\phi/\rho^*)}$$

for all $\phi > 0$.

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Enveloping

$$\mathbb{P}(-\tilde{z} > \phi) \leq \frac{1}{u(\phi/\rho^*)}$$

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- A small ρ^* means a tight probability bound at all loss levels.

- Uncertain constraints
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Utility-based CREM

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Example

Suppose $\rho^* = 1$ and $u(x) = \exp(x)$. Then we have the following bounds.

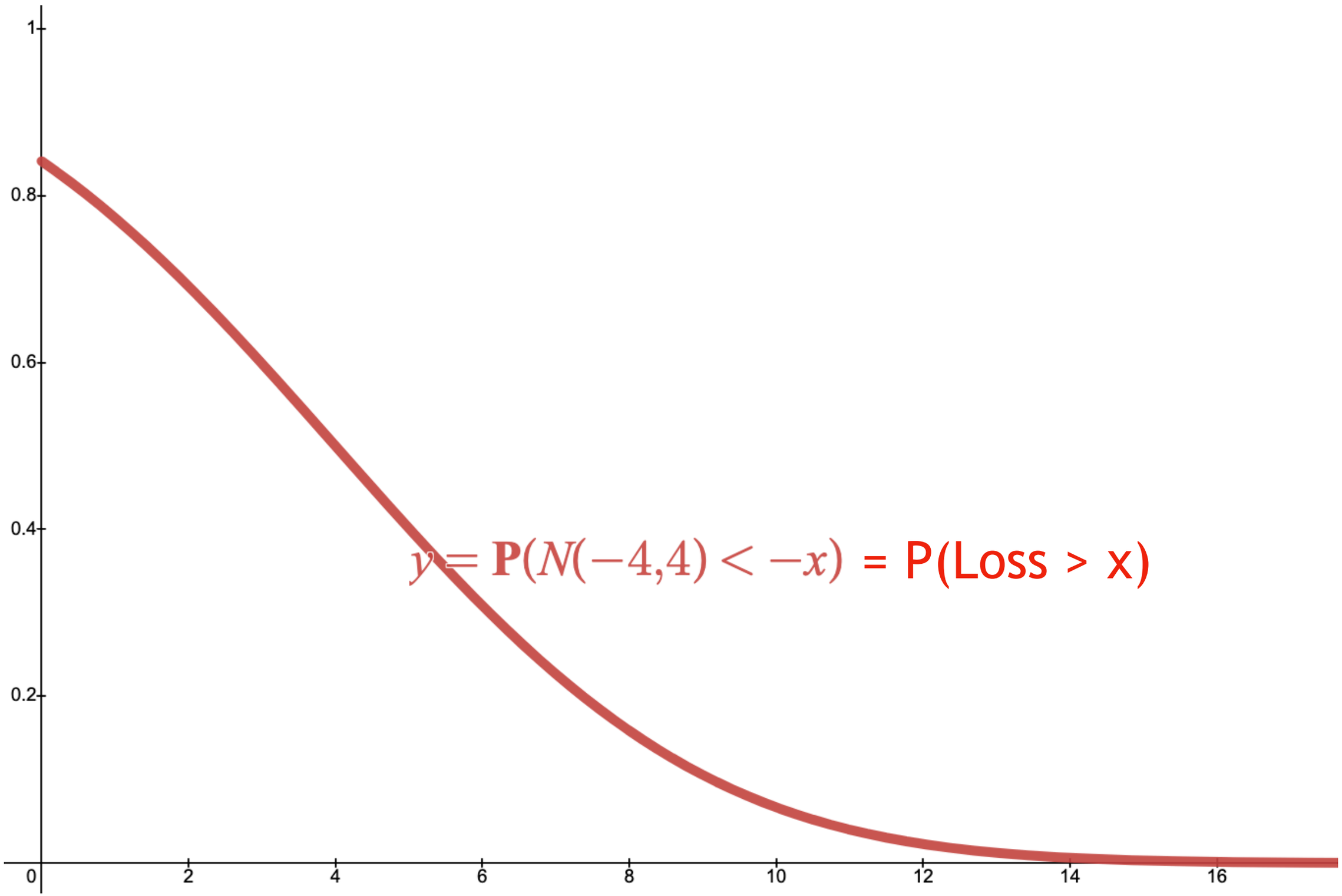
P(Loss > 1)= $\mathbb{P}(-\tilde{z} > 1) \leq 0.36$

$\mathbb{P}(-\tilde{z} > 2) \leq 0.13$

$\mathbb{P}(-\tilde{z} > 3) \leq 0.05$

$\vdots$

A Numerical Example



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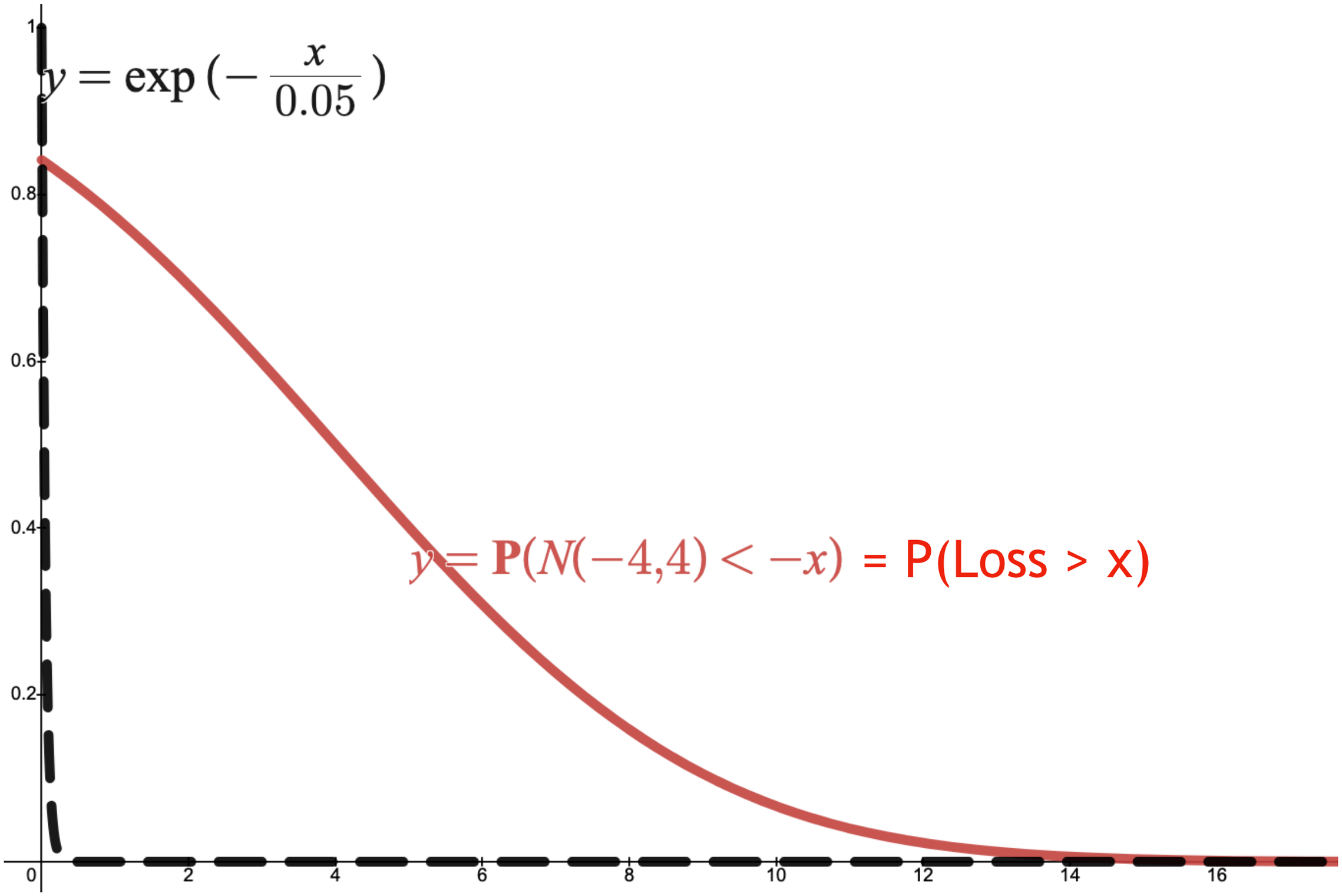

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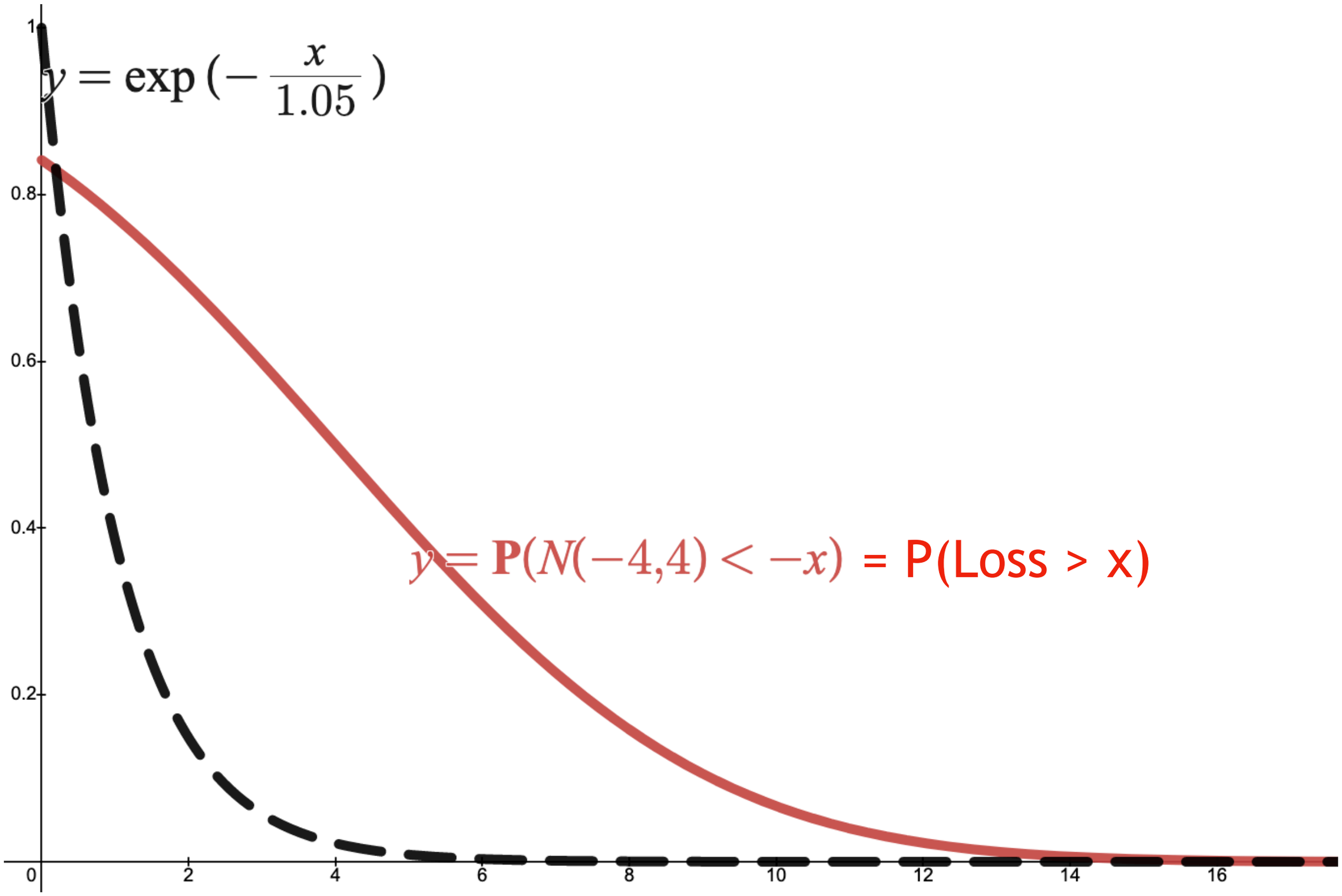

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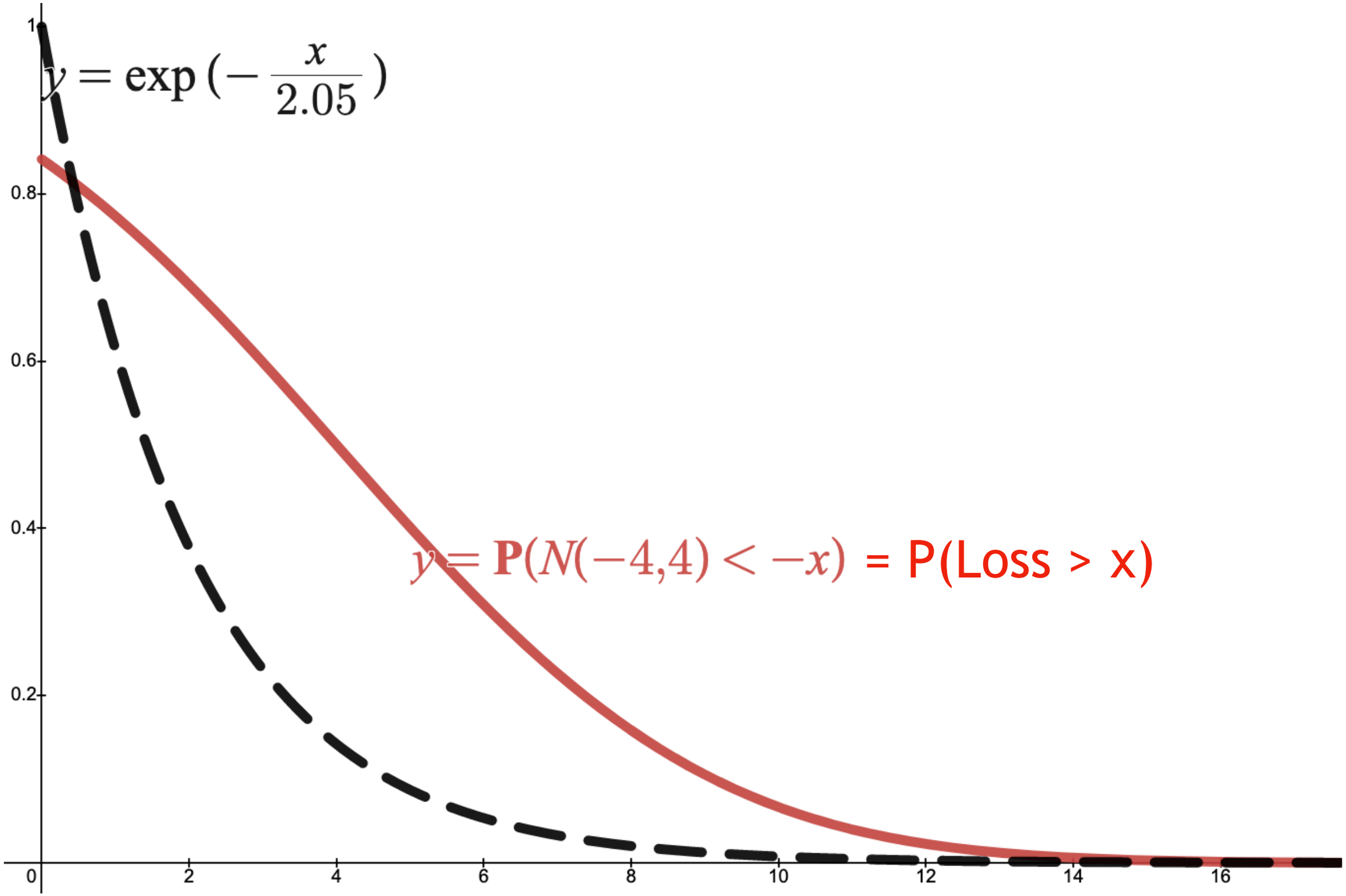

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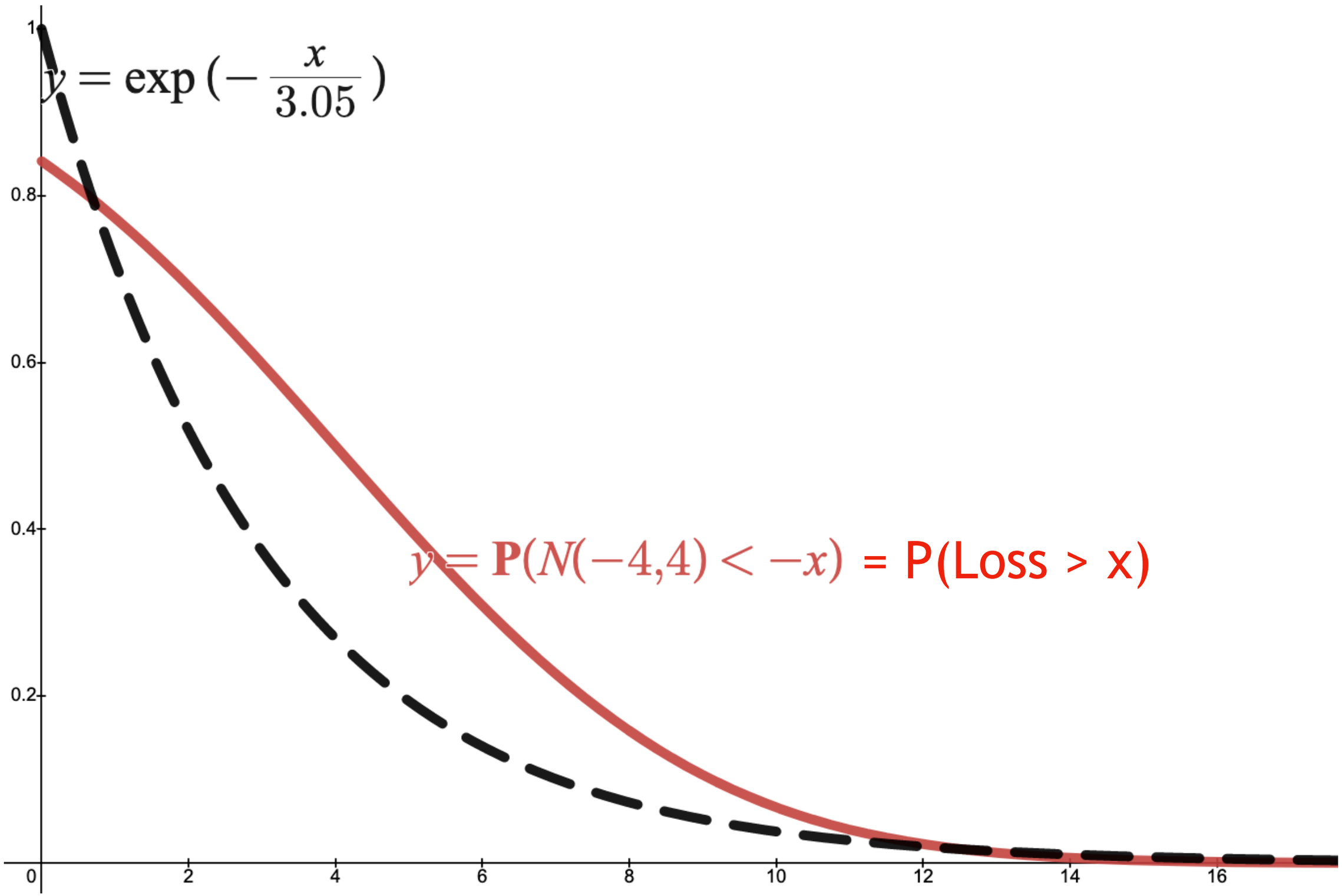

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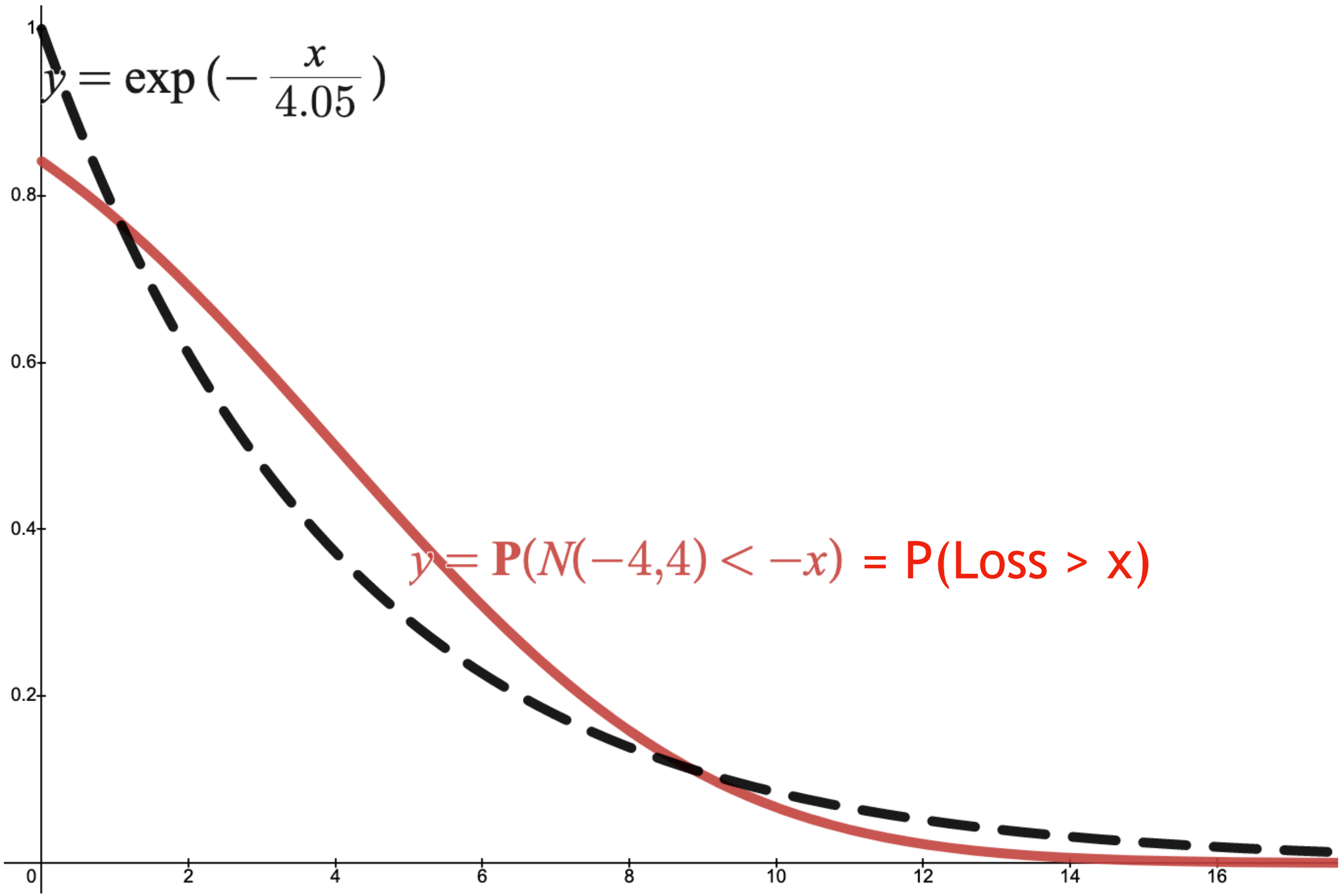

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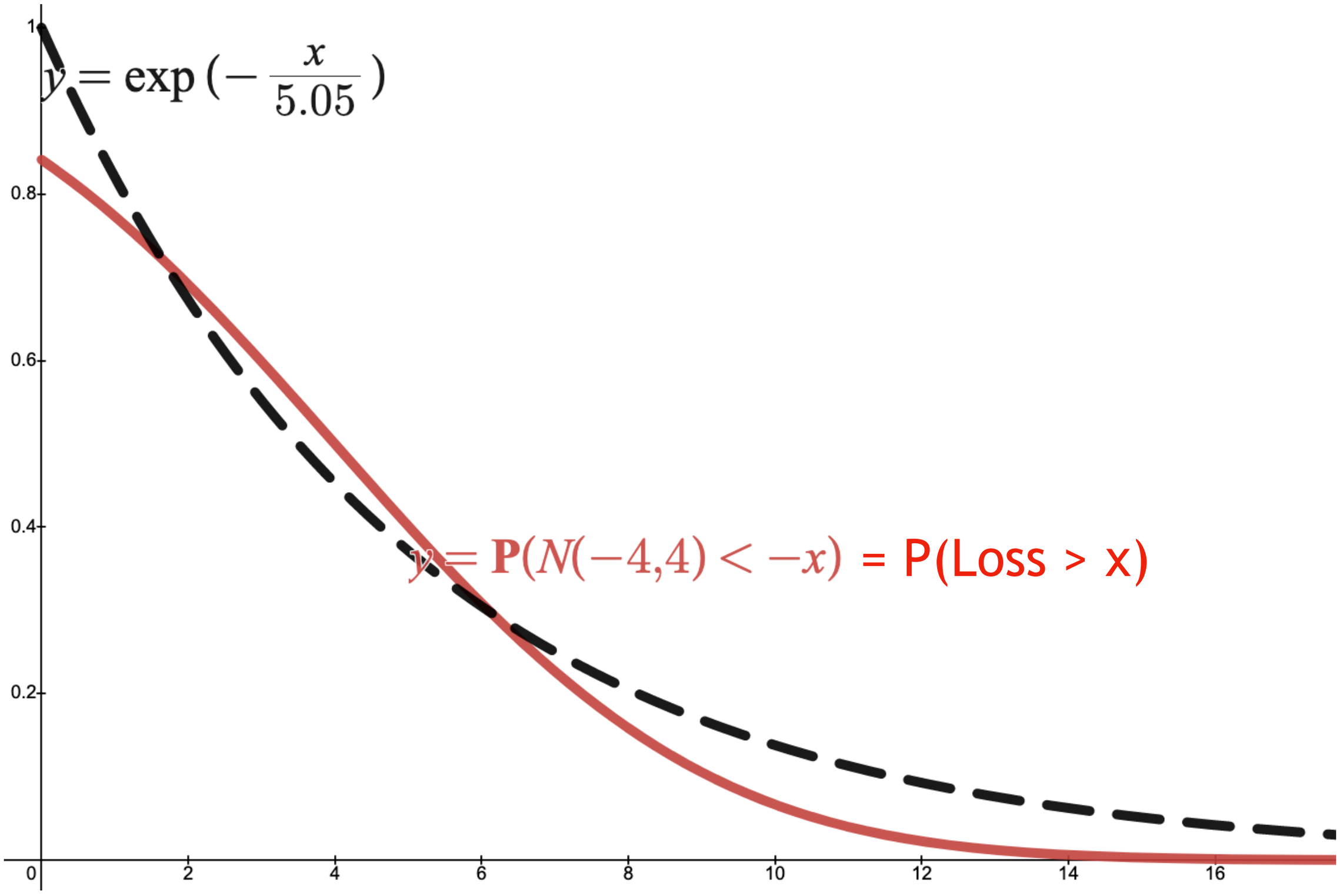

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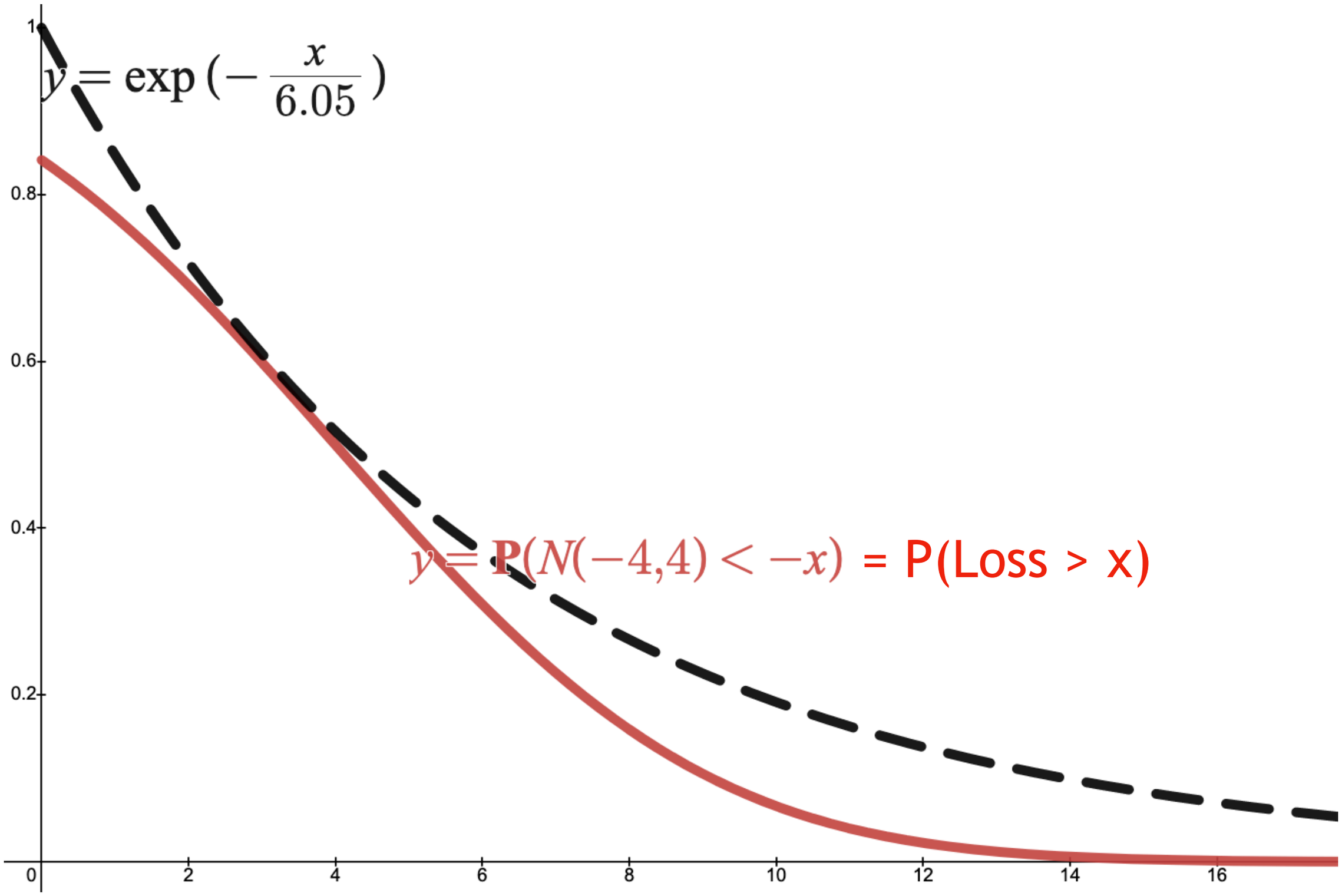

A Toy Example

Chance-constrained

$$
\begin{aligned}
\min\ & \text{Total Cost} \\
\text{s.t.}\ & \mathbb{P}(\text{Overtime} \geq 5) \leq 0.10 \\
& \mathbb{P}(\text{Overtime} \geq 10) \leq 0.05 \\
& \mathbb{P}(\text{Overtime} \geq 20) \leq 0.01 \\
& \vdots \\
& x \in \mathcal{X}
\end{aligned}
$$

A Toy Example

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Utility-based CREM

$$\begin{aligned} \min\ & k \\ \text{s.t.}\ & \mathbb{E}\left[u\left(-\frac{-\text{Overtime}}{k}\right)\right] \leq 1 \\ & x \in \mathcal{X} \\ & k > 0 \end{aligned}$$

More in the paper

Other Important Results

Ambiguity Set	Utility Function	Decision Variables
General Moment	General	Continuous
General Moment	Piecewise Linear	Continuous
Mean-covariance	Piecewise Linear	Continuous
Mean-covariance	Piecewise Linear	Integer

Surgery Allocations

- We need to assign I surgery cases to J operating rooms.
  - The duration of a surgery case is random.

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- Costs:
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- Two conflicting goals:
  - Minimizing the overtimes (OT)
  - Minimizing the opening and assignment costs

Surgery Allocations

- Zhang et al. (2018) considered the distributionally robust version of this surgery allocation problem.

Reference:

Zhang, Y., Jiang, R., Shen, S., “Ambiguous chance-constrained binary programs under mean-covariance information,” SIAM Journal on Optimization, 28(4), 2922-2944, 2018.

Formulations Used

Chance-constrained

$$C^* = \min \text{ Total Cost}$$
$$\text{s.t. } x \in \mathcal{X}$$
$$\sup_{\mathbb{P} \in \mathcal{P}} \mathbb{P}(\text{Room } i \text{ has OT}) \leq \alpha \quad \text{for all } i$$

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Utility-based CREM

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\begin{aligned}
\min\ & k \\
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& \sup_{\mathbb{P}\in\mathcal{P}} \mathbb{E}\left[u\left(-\frac{-\text{OT}_i}{k}\right)\right] \leq 1 \quad \text{for all } i \\
& k > 0 \\
& \text{Total Cost} \leq C^*
\end{aligned}
$$

More in the paper

Results at a Glance

Prob.	Approach	Avg.	Std.	Worst Case	Solving Time (s)
0.3	Utility-based CREM	0.47	4.29	177.50	2
0.3	Chance Constraint	0.68	5.82	215.75	57
0.5	Utility-based CREM	3.15	12.96	257.29	2
0.5	Chance Constraint	3.41	14.55	275.17	43

Smaller is better

Conclusion

Chance-constrained

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\begin{aligned}
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Conclusion

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